Housing First or Treatment First?

A stylized model

August 15, 2026

1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a substance use disorder, and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind, and it has six state variables and roughly forty parameters, which is what the welfare and fiscal questions require but makes it hard to see which piece of the environment does the work. This note strips the environment to the smallest one in which three questions are still well posed.

1. **Who uses drugs?** Use is concentrated at the bottom of the income distribution. Drugs are a normal good bought at a price, so that concentration is not what a model produces by default. Which feature of the environment reverses it, and can any other?
2. **What does housing assistance do to that decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. It also cushions the consequence of acquiring the disorder, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In the model as it stands, never. There are no externalities from addiction and homelessness, and so Housing First always raises welfare when agents are forward looking, as they are here. Addiction creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension alone cannot justify conditionality. However, if addiction and homelessness generate an externality, such as reducing the earning potential of children, then Treatment First may be preferred. The model's work here is to show why the first claim holds and then to price the externality that overturns it.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component has been removed and the model refitted without it, so what remains is what the results need. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question.

2 The environment

2.1 Households and preferences

There are three periods and a continuum of households. A period stands for roughly twenty years of adult life. Survival, recovery and the unemployment probabilities are per period numbers, while β and R take conventional annual values rather than compounded ones, deliberately: compounding over twenty years would leave the continuation value too small to matter, and everything in Section 4 runs through the continuation value. Nothing below turns on the exact horizon, since every comparison varies one object and holds the rest of the parameter vector fixed.

A household is characterised by its permanent income y, fixed over its life and lognormally distributed, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma^2, \sigma^2)$, so mean income is one and every quantity below measured in units of income is a multiple of the population average. Households differ in y and in nothing else, so every household at a given income makes the same choice and drug use is a threshold in income rather than a probability. That also keeps intact the paper's claim to generate the income gradient without ex ante preference heterogeneity: there is none here to do the work.

Flow utility depends on non durable consumption c, the flow of housing services h consumed that period, and drug consumption v,

$$u(c,h,v) \;=\; (1-\gamma-\theta)\log c \;+\; \gamma\log h \;+\; \theta\big[\log(b+v)-\log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so γ is the housing budget share proper. Housing services are chosen each period from the three point grid $\{h_0, h_S, h_L\}$ of equation (5). The pair (θ, b) governs the drug term: the marginal utility of the vth unit is $\theta/(b+v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone and Geary form keeps the first unit's marginal utility finite, since under plain $\theta\log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it: it gives mortality a cost that does not vanish when consumption is low, and Section 4 turns on it. Households discount at β, save in one risk free asset at gross return R, and cannot borrow.

2.2 Timing

Period 1. Every household is employed and clean. It chooses consumption, housing, drug use $v \geq 0$ and saving, paying market rent on whatever unit it picks. For simplicity, this is the only period in which it chooses whether to use. Using generates the risk of a substance use disorder next period, rising in v,

$$\pi(v) \;=\; 1-e^{-\lambda v}. \tag{2}$$

Because π is increasing, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

Period 2. For households that used, a disorder $d \in \{0,1\}$ is realised with probability $\pi(v)$; for the rest it is not. Employment status $\chi \in \{0,1\}$ is realised, with $\chi = 1$ denoting unemployment, which occurs with probability u_0, or the higher u_1 if $d = 1$. An unemployed household receives income support. It then chooses consumption, housing and saving. The probability of recovering by the next period is ρ.

Period 3. Terminal. Employment, income support, the voucher and the housing choice work as in period 2, there is no drug use decision, and assets are run down with nothing bequeathed.

Mortality, and why there are three periods. A household survives from one period to the next with probability

$$\phi(d) \;=\; \phi_0\,(1-\kappa d), \tag{3}$$

so κ is the proportional reduction in survival caused by the disorder. The third period is what makes that reduction bite. Mortality is a tax on a continuation value, so it can only matter if there is a period after the one in which the disorder is carried. In a two period version the disorder is realised in the last period, death costs nothing because nothing follows it, and the mechanism the whole model rests on is silent: recalibrated as a two period model, the moral hazard of Section 5 collapses from +3.57 percentage points to +0.23. Three periods is the smallest number at which acquiring the disorder has a future to destroy.

2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the disorder: there is no wage penalty conditional on working, because the quantitative model sets that parameter to one. Income in a period, given permanent income y and employment status χ, is

$$y_e(y,\chi) \;=\; \begin{cases} y & \text{if } \chi = 0, \\ \psi\cdot\min\{y,\bar{y}\} & \text{if } \chi = 1, \end{cases} \tag{4}$$

with ψ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The quantitative model has persistent productivity shocks as well, and they are not a key channel for anything asked here:

what links the disorder to housing is the loss of a job, not the ordinary fluctuation of earnings around a trend.

The level of ψ links the disorder to homelessness: for the poorest households half of income falls short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income, and were income support above that rent the moral hazard of Section 5 would be zero. The cap matters instead for Section 4: without it the resource cost of the disorder would be exactly proportional to income, and a penalty proportional to income cannot move a gradient that income itself generates.

Consumption is the numeraire, and the rent per unit of housing services p_r is normalised to one, so a unit's size and its market rent are the same number. Each period the household chooses

$$h \in \{h_0,\ h_S,\ h_L\}, \qquad h_0 < h_S < h_L, \tag{5}$$

homelessness, a small unit and a standard unit. Homelessness is free and delivers almost nothing. Two rented sizes is the fewest that lets a household economise on housing rather than lose it: with one, homelessness comes out at 3.29 per cent against a target of 0.40, at any h_0.

A voucher covers part of the rent, is means tested and is rationed. Writing $z \in \{0, 1\}$ for whether one is held,

$$\Pr(z = 1 \mid y_e, d) \ = \ \pi_V \cdot \mathbf{1}\{y_e \le y_V\} \cdot \mathbf{1}\{\text{policy covers } d\}, \tag{6}$$

and a household holding one pays the fraction κ_r of market rent, so the rent paid on option j is

$$r_j(z) \ = \ p_r\, h_j \big[\kappa_r z + (1 - z)\big] \cdot \mathbf{1}\{j \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in periods 2 and 3. The parameter π_V is the generosity of the programme and is the policy dial of Section 5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of d. **Treatment First** covers only households without the disorder. Since cover in the disorder state is the only difference, the gap between them is the moral hazard and the whole of it.

2.4 The household's problem

Solve backward, with a assets carried in, y permanent income, χ employment status, z the voucher and d the disorder, and write $c_j = Ra + y_e(y, \chi) - r_j(z) - a'$ for consumption on housing option j. In period 3 the household consumes everything, so $a' = 0$, and chooses housing,

$$\widetilde{W}_3(a, y, \chi, z) = \max_j \big\{(1 - \gamma - \theta) \log c_j + \gamma \log h_j + \aleph\big\}, \tag{8}$$

and the value of entering period 3, taking expectations over employment and the voucher lottery, both of which depend on d, is $W_3(a, y, d) = \mathbb{E}[\widetilde{W}_3 \mid y, d]$. In period 2 the household also saves and faces mortality and recovery,

$$\widetilde{W}_2(a, y, \chi, z, d) = \max_{j,\, a' \ge 0} \big\{(1 - \gamma - \theta) \log c_j + \gamma \log h_j + \aleph + \beta\, \phi(d)\, \mathbb{E}_{d'} W_3(a', y, d')\big\}, \tag{9}$$

with $W_2(a, y, d) = \mathbb{E}[\widetilde{W}_2 \mid y, d]$ and $d' = 0$ with probability ρ if $d = 1$ and $d' = d$ otherwise. In period 1 the household is employed, clean and holds no voucher, so $c_j = y - r_j(0) - p_v v - a'$, and it chooses housing, drug use and saving,

$$V_1(y) = \max_{j,\, v \ge 0,\, a' \ge 0} \big\{(1 - \gamma - \theta) \log c_j + \gamma \log h_j + \theta[\log(b + v) - \log b] + \aleph + \beta\, \phi_0\, \mathbb{E}_{d'} W_2(a', y, d')\big\}, \tag{10}$$

where the disorder arrives with the use dependent probability of (2). Housing is a choice in period 1 as in every other period, which is why $r_j(0)$ appears there: y is income, not income net of an assumed rent. A household uses if the value at its best positive v exceeds the value at $v = 0$, and because households differ only in y that defines a set of income intervals on which use occurs.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor	Standard	0.96
R	Gross interest rate	Standard	1.03
σ	Dispersion of log permanent income	Heathcote, Perri and Violante (2010)	0.75
ϕ_0	Survival per period, no disorder	CDC life tables	0.93
$\aleph$	Flow value of being alive	Viscusi and Aldy (2003)	2.80
γ	Weight on housing services	Consumer Expenditure Survey	0.22
θ	Weight on drug use	Set; b is calibrated given it	0.11
The disorder			
ρ	Recovery probability per period	SAMHSA, TEDS	0.40
κ	Proportional fall in survival with the disorder	Hjorthøj et al. (2023)	0.45
u_0	Unemployment probability, no disorder	BLS	0.049
u_1	Unemployment probability, with the disorder	NSDUH 2024, ages 26 and over, weighted	0.1625
Income support			
ψ	Replacement rate while unemployed	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	DOL, state benefit maxima	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Numeraire	1.00
p_v	Price of a unit of drugs	Normalisation of the units of v	1.00
h_L	Standard rental unit	ACS 2023, rent over income	0.212
h_S	Small rental unit	Half the standard unit	0.106
κ_r	Tenant's share of the rent when covered	HUD voucher rules	0.30
y_V	Voucher means test, multiple of average income	HUD income limits	1.00
π_V	Voucher generosity in the baseline	HUD, share of eligible assisted	0.27

3 Calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: β and R take standard values, the dispersion of log permanent income is $\sigma = 0.75$ from Heathcote, Perri and Violante (2010), per period survival is $\phi_0 = 0.93$ from the CDC life tables, and $\aleph$ puts the implied willingness to pay for survival in the range of the value of a statistical life estimates surveyed by Viscusi and Aldy (2003). The housing weight γ is a quarter of non drug spending, as in the Consumer Expenditure Survey, and θ is set rather than calibrated, since it is b that governs whether anyone uses at all. For the disorder, recovery over a period is $\rho = 0.40$ from the SAMHSA treatment episode data, the proportional fall in survival is $\kappa = 0.45$ from Hjorthøj et al. (2023), and the two unemployment probabilities are measured directly, $u_0 = 0.049$ from the BLS and $u_1 = 0.1625$ from NSDUH 2024 for respondents aged 26 and over. Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The standard rental unit costs 21.2 per cent of mean income, its rent to income ratio in the ACS, and the small unit is half of it. The voucher parameters are institutional: covered tenants pay 30 per cent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the HUD share of eligible households actually assisted.

Three parameters are set inside the model, in Table 2. Each is bracketed on one moment in sequence rather than fitted jointly: with no preference heterogeneity the aggregate is a cliff in b, so a gradient method steps out of the region where participation is interior and walks to a corner. The pull of the first unit of drugs b is matched to a drug participation rate of 5.13 per cent, housing services on the street h_0 to a homelessness rate of 0.399 per cent, and the addiction hazard λ to the 33.2 per cent of users who acquire a disorder, from the NSDUH and the HUD point in time counts. Participation is bracketed last, because all three parameters move it and only the moment fitted last is hit exactly.

4 Who uses drugs

Start from the force any explanation has to overcome. Drugs are a normal good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00019
h_0	Housing services when homeless	Homelessness rate	0.399%	0.432%	0.00070
λ	Addiction hazard per unit of use	Share of users who acquire the disorder	33.2%	29.7%	950.5

Note: one untargeted moment is worth recording. The share of the homeless carrying the disorder comes out at 0.42, close to the data, and it is not a free number: setting u_0 to zero makes every route into homelessness run through the disorder and takes that share to 0.997.

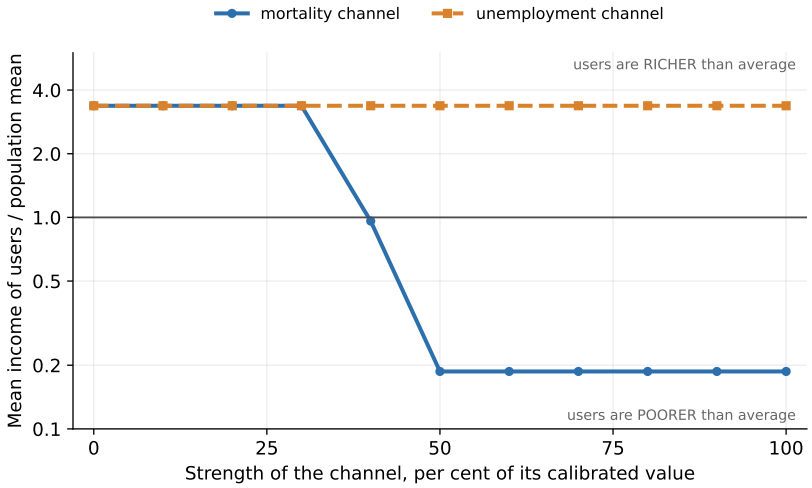


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and b is recalibrated at every point so that participation is 5.13 per cent throughout.

for acquiring a disorder that is the only force operating, and drug use concentrates at the top: the average user earns 3.37 times the population average. Two features of the environment could reverse that. The *mortality channel* is that the disorder kills you, through κ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home.

Figure 1 scales each from nothing to its calibrated value, recalibrating b at every point so participation stays at 5.13 per cent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

The result is sharper than the statement that mortality matters more. That channel reverses the gradient at half of its calibrated strength, taking the income ratio from 3.37 to 0.19, and does not move again; the unemployment channel does not reverse it at all, leaving users at 3.37 times average income even fully switched on. A resource penalty *cannot* produce the concentration of drug use at the bottom, however strong it is made.

The asymmetry is not obvious, and the same logic operates in the quantitative model. A resource penalty scales with the very thing it is meant to offset: taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. A mortality penalty is different in kind, because what it destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. This is why $\aleph$ appears in equation (1): it is the part of the continuation value that does not shrink as consumption falls. The same channel carries the policy result of Section 5. Switch mortality off and the deterrent effect of a universal voucher falls from -3.00 percentage points to -0.37, which is to say it disappears: a voucher deters nobody once what drug use threatens is only

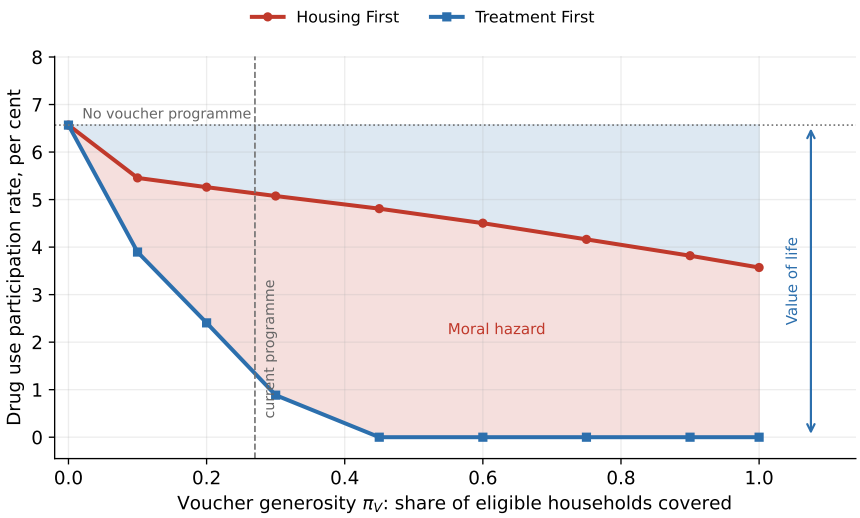


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the current programme, $\pi_V = 0.27$.

resources, because resources are what the voucher supplies either way.

5 What housing assistance does

The two forces of question 2 can be separated exactly, because they differ in what happens on acquiring the disorder. Under Treatment First the voucher is withdrawn at that point, so only the value of life force operates; under Housing First it is not, so both do. Comparing the two policies at the same generosity therefore isolates the moral hazard, and comparing Treatment First with no programme at all isolates the value of life.

Figure 2 traces both as generosity rises. With no programme, 6.57 per cent of households use. Universal Treatment First takes that to nothing, a value of life force of -6.57 percentage points; universal Housing First leaves it at 3.57 per cent, so the moral hazard is $+3.57$ points and the net effect of Housing First is -3.00. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim while sounding as though they are making a theoretical one.

Drug use is not what a housing programme is for, and a reader shown only Figure 2 could conclude Treatment First wins on every count. Figure 3 plots the outcomes the programme is aimed at alongside drug use. On homelessness the two policies end in the same place, both driving it from 0.63 per cent to zero, with a gap along the way never exceeding a ninth of a point. The difference is on the other margin, where at universal cover Housing First leaves 0.63 per cent of the population carrying the disorder and Treatment First leaves none: the moral hazard shows up as disorder cases among households who are housed, not as families left on the street.

6 Welfare

6.1 Without externalities, Housing First always wins

Compare the policies at the same π_V. They differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger

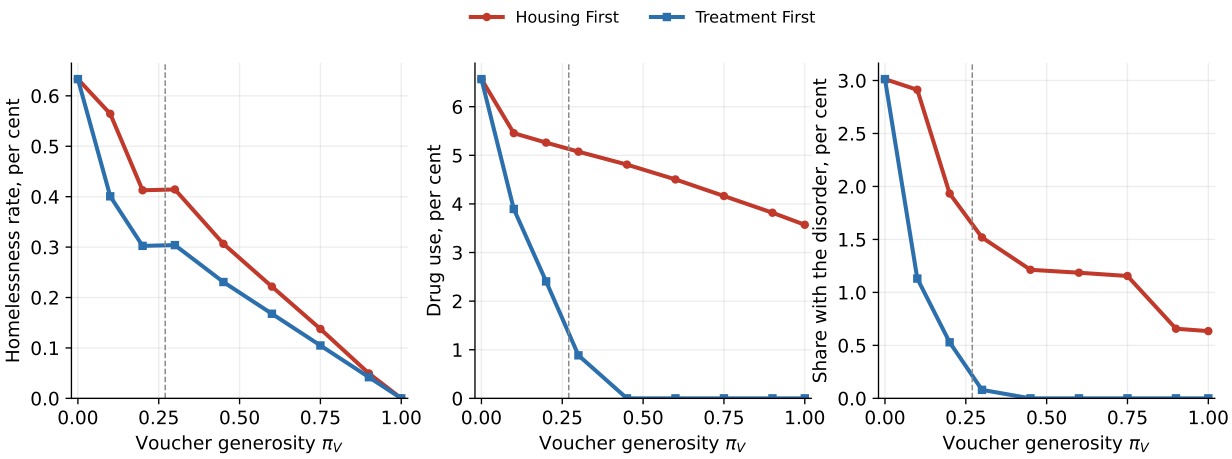


Figure 3: Homelessness, drug use and the disorder under the two policies

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	3.57%	0.00%
Share with the disorder	0.635%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.1830	0.1833
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		4.1%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.39%
Consumption equivalent loss, whole population		0.016%

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household from period 2 on, discounted, per unit of average income.

in every state and strictly larger in one, so a household that optimises cannot be worse off:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{11}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly, by not using. Conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.

It follows that the moral hazard is not a distortion. Households here face the true addiction hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household taking a risk it is now better able to bear, as an insured household drives in worse weather. The +3.57 points of Section 5 is a statement about drug use and, on its own, none about welfare. Table 3 puts numbers on it: 4.1 per cent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.39 per cent of consumption.

A fixed budget does not rescue conditionality either. Treatment First is slightly *more* expensive at universal cover, 0.1833 against 0.1830, so matching on cost rather than on generosity changes nothing. The reason is unattractive and true: survival to period 3 falls from 0.93 to 0.51 with the disorder, so the households the moral hazard adds to the rolls die sooner and stop costing money.

So a case for conditionality must come from outside the model as it stands. Households make no mistakes, so there is no internality; the policies cost the same, so there is no fiscal argument; and drug use here touches

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's disorder rate	Ratio	Loss to the child
0.00	1.80%	1.00	0.00
0.20	3.23%	1.80	0.59
0.30 (calibrated)	4.37%	2.43	1.01
0.40	5.95%	3.31	1.58
Panel B. Does the externality reverse the ranking?			
Private welfare loss from conditionality			0.044%
Disorder cases prevented			0.635 pp
Damage to the child of one such case			90%
Externality gain from conditionality			0.570%
Gain over loss			13.0

Note: Panel A is computed at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; loss to the child is its consumption equivalent loss per lifetime income. Panel B is at universal cover, one child per household, no discounting across generations.

nobody but the user. The last is the one that can be measured rather than assumed, and the table says in advance how large it must be: conditionality prevents 0.635 percentage points of disorder cases at a private cost of 0.044 per cent of one lifetime income, so a social cost of 0.069 of a lifetime income per case would be exactly enough to justify it.

6.2 The externality on children reverses it

The link from a parent's addiction to a child has to run through income rather than through preferences. Letting the child of a user inherit a stronger taste, a lower b, matches the transmission fact but has the wrong welfare sign: b enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. Letting it inherit a greater susceptibility, a higher λ, fails the other way, since a child facing a riskier drug uses *less*. Income works on both counts, and it works because of Section 4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use. The income gradient does the whole job, so the model transmits addiction across generations *without transmitting any preference for drugs*. A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the disorder}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{12}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into V_1. General income persistence is absent because it is not an externality: it operates whether or not the parent uses.

The two penalties are set to different things. Panel A of Table 4 shows that $\delta_d = 0.30$ makes the child of a parent who acquired the disorder 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. For δ_h, Chetty, Hendren and Katz (2016) find children moved to a low poverty neighbourhood before age 13 earned 31 per cent more as adults, which is conservative here since growing up homeless is presumably worse. The two do different work: both policies drive homelessness to zero at universal cover, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality is 0.044 per cent of a lifetime income, small because only 4.1 per cent of households are affected. Against that, conditionality prevents 0.635 percentage points of disorder cases, and each costs one child 90 per cent of a lifetime income, far above the 6.9 per cent threshold set out above. The externality gain is thirteen times the private loss, and stays above it throughout the range of δ_d in Panel A. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

Whether any of this is an externality at all depends on how much of it parents already take into account, and the evidence points both ways. Substance use falls sharply during pregnancy in the NSDUH and rises

again afterwards, and entry into treatment clusters around child welfare involvement, neither of which fits a parent placing no weight on a child. Equally plainly the weight is not one: parental drug use is recorded in roughly a third of foster care removals, and Altonji, Hayashi and Kotlikoff (1992) reject full altruism outright. A pure generational discount is a useful anchor for how much to allow, and at $\beta = 0.96$ over thirty years it is 0.29. This is the assumption the case for conditionality is most sensitive to, and the quantitative model treats it as such, putting the child's value in the parent's own objective at a weight of 25 per cent rather than assuming the weight is zero.

6.3 What this settles and what it does not

One feature of the model inflates that margin, and it is the right note on which to hand the question over. Treatment First drives drug use to *exactly* zero here, an artifact of having no preference heterogeneity: a policy strong enough to deter the marginal user deters every one. Nobody faces the condition and fails to meet it, and those households are the entire empirical case for Housing First. Establishing which way each comparison goes, with them, a government budget, an equilibrium drug price and a proper life cycle, is what the quantitative model is for.

References

Chetty, R., N. Hendren and L. Katz (2016). "The Effects of Exposure to Better Neighborhoods on Children." *AER* 106(4). Altonji, J., F. Hayashi and L. Kotlikoff (1992). "Is the Extended Family Altruistically Linked?" *AER* 82(5), 1177–98. Heathcote, J., F. Perri and G. Violante (2010). "Unequal We Stand." *RED* 13(1), 15–51. Hjorthøj, C. et al. (2023). Excess mortality among people with substance use disorders. McGovern, R. et al. (2023). "Maternal and paternal substance use and child substance use." *Addiction* 118(5). Viscusi, W. K. and J. Aldy (2003). "The Value of a Statistical Life." *Journal of Risk and Uncertainty* 27(1), 5–76. Data: SAMHSA (NSDUH 2024, TEDS); HUD; BLS; ACS; CDC; DOL.